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# How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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## Abstract

What shapes the content of a recurrent navigation agent’s learned internal state — the task, the architecture, or the structure of the visual input? Prior work has shown that blind PointNav agents spontaneously develop cognitive-map-like representations in their recurrent state under task pressure alone [Wijmans et al., 2023]. We show that the blind result is a special case of a broader principle: *the visual encoder’s information capacity determines whether the LSTM has to carry world-frame spatial structure*. Holding task, optimiser, and LSTM backbone identical and varying only the visual pathway (no vision; a resolution-collapsed encoder; uniform vision; foveated vision at a fixed gaze; foveated vision with a learned gaze), we probe the recurrent state for world-frame position and heading. When the encoder cannot resolve those quantities, the LSTM develops a persistent, integration-capable code for them; when it can, the recurrent state drifts to pass-through. We call this the *encoder–memory race*; it recasts the “cognitive maps emerge without vision” story as “cognitive maps emerge when the visual pathway cannot resolve the map on its own.” A second finding cuts across conditions: five agents that solve the task equally well develop recurrent representations in essentially orthogonal linear subspaces, and cross-condition memory transplants incur task cost well above transplant mechanics — sensor structure shapes representational *format*, not only strength. A third question — whether static gaze location within a rich-encoder regime acts as a second content axis — is tested by a foveated-shifted causal control whose training is in progress. Three implications. (i) Encoder capacity is a controllable lever for inducing cognitive-map-like internal state, not a passive feature of a given vision system. (ii) “More pixels, richer memory” is the wrong intuition; the relationship is inverse. (iii) The pattern parallels independent animal-navigation findings — enhanced hippocampal coupling in congenitally blind humans, the disruption of rodent grid-cell firing in the dark (with navigation nonetheless preserved), cross-modality hippocampal remapping in bats — without depending on their biophysics, supporting a sensor–memory coupling principle that is not specific to LSTMs.

## 1 Introduction

Recurrent navigation agents trained with deep RL build internal states that support memory, planning, and generalisation, and the content of those internal states is at least as interesting as the task metrics they drive. A natural question is what *determines* that content: architecture, objective, training budget, or visual input? Wijmans et al. [Wijmans et al., 2023] showed that map-like spatial representations emerge from task pressure alone, even in blind agents that receive only goal displacement and proprioception; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] demonstrated grid-cell-like structure in path-integration networks. Together these results establish that *some* spatial structure is learned without being designed in. They leave open the complementary question that we take up

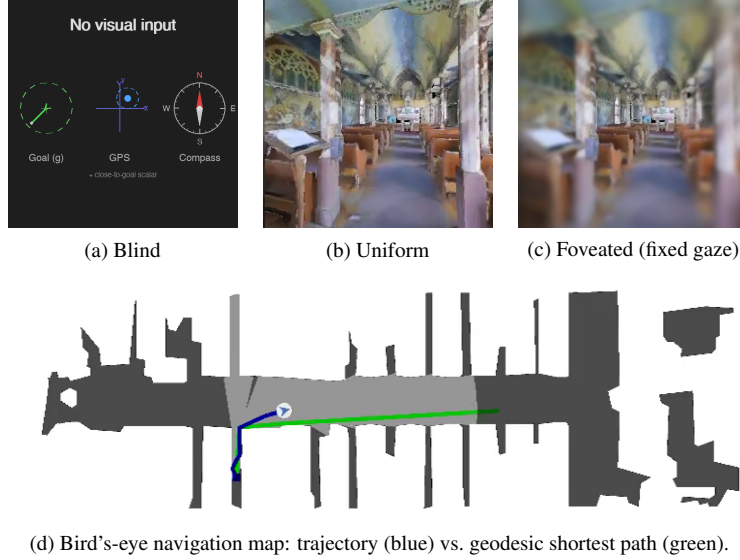


Figure 1: The five visual conditions differ only in what reaches the agent’s encoder. (a) Blind: no visual input; only the non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar) — the Wijmans [Wijmans et al., 2023] configuration. (b) Uniform:  $256 \times 256$  full-resolution RGB. (c) Foveated (fixed gaze): eccentricity-dependent Gaussian blur centred on the image middle ( $\sigma_{\max} = 8$ , quadratic falloff). The two additional conditions not shown are *matched-compute* (a down-sampled uniform view with the same total pixel budget as foveated) and *foveated (learned gaze)*, which adds a learned  $(x, y)$  gaze offset to (c). (d) All conditions solve PointGoal navigation in the same Habitat Matterport3D / Gibson scenes; the top-down view is shown for reference only and is not available to the agent.

here: within agents that all achieve high task success, how much of *what* recurrent memory encodes is determined by the structure of the visual input rather than by the task? In biology the answer is clear — primate hippocampal cells code joint place–view–action state because primates sample the scene with a foveated sensor [Wirth et al., 2017, Rolls, 2024], and the same bat produces non-overlapping hippocampal populations in vision versus echolocation [Geva-Sagiv et al., 2016]. We investigate the analogous question in a controlled agent setting where the only free variables are the visual input structure and the gaze that directs it.

Our five-condition design isolates three degrees of freedom: sensor type (blind / uniform / foveated), information volume (uniform vs. matched-compute), and gaze location (fixed vs. learned). Every condition uses the same DD-PPO learner, the same Gibson-0+MP3D training pool, the same LSTM backbone, and the same non-visual sensor stack (Figure 1). We adopt the LSTM architecture from Wijmans et al. [Wijmans et al., 2023] deliberately: the emergent-map literature we build on is recurrent, and matching that architecture is what lets our findings be interpretable as statements about *the structure of the task-shaped memory* rather than about the recurrent-versus-transformer design choice. We make our claims specifically about recurrent PointNav agents, and discuss in §5 whether the logic plausibly extends to transformer-backed or non-navigation systems (an empirical question we do not settle here). Within this setup, two accounts make divergent predictions for what a foveated agent’s memory should do. A *passive buffer* stores what it currently perceives, so decodable content should fall when current input is degraded. A *compensatory* memory invests in internal state when current input is unreliable, so decodable content should persist over longer horizons than for an agent whose current input is always reliable. Which pattern the agent converges to, and whether gaze location can act as a separate representational-content lever even when task performance is held fixed, are the questions this paper asks empirically.

We test three hypotheses. **H1 (visual-encoder bottleneck  $\rightarrow$  LSTM spatial compensation)**. Agents whose visual encoder cannot resolve world-frame position or heading on its own should develop these quantities in their recurrent state; agents with a rich visual encoder should not. Signature: current-state GPS/compass probe  $R^2$  tracks visual-encoder information content monotonically across five

conditions (blind, matched-compute at  $1 \times 1$  feature map, uniform, foveated, foveated-with-learned-gaze), with a persistent-integration signature (decoding remains accurate at lags of 5–10 time steps). **H2 (sensor structure  $\rightarrow$  format-level divergence).** Different sensor conditions should produce internal representations in distinct linear subspaces, not scaled versions of a common code. Signature: pairwise CKA near zero, probe weights fail to transfer across conditions (while within-condition probes succeed), and cross-condition memory transplants degrade task performance beyond same-condition self-transplants. **H3 (gaze location  $\rightarrow$  orthogonal content lever).** Where a foveated sensor is centred (*gaze location*) should act as a representational-content axis independent of the sensor transform. We test this with a foveated-shifted control in which gaze is hardcoded to a different image location from the default central gaze; same foveation transform, same training budget, different gaze target. The behavioural effect of shifting gaze location on the LSTM’s probe signature is what we report (§4.4). Our minimal end-to-end learned-gaze module (a deterministic sigmoid MLP with slow-gaze updates, §4.4 and Appendix B) collapsed to a static output; we use that collapsed location as the foveated-shifted control rather than as evidence about active gaze. Each hypothesis has a direct analog in the animal-navigation literature [Kupers and Ptito, 2014, Chen et al., 2016, Geva-Sagiv et al., 2016, Jutras et al., 2013, Najemnik and Geisler, 2005], which we return to in Discussion as evidence that the effects reflect general sensor–memory coupling principles rather than architectural artefacts.

**Findings** H1 is supported by a clean monotonic effect: LSTM world-frame GPS encoding tracks visual-encoder information bottleneck from  $R^2 = 0.95 \pm 0.02$  (blind, no encoder) and  $R^2 = 0.78 \pm 0.10$  (matched-compute,  $1 \times 1$  encoder) down to chance ( $R^2 \in [-2.4, +0.06]$  with CV  $\sigma \in [0.7, 4.0]$ ) for the three full-resolution-encoder conditions (uniform, foveated, foveated-learned). H2 (representational-format divergence) is supported independently by near-zero cross-condition CKA and by mid-episode memory-transplant SPL drops of 0.19–0.21. H3 asks whether gaze location is a second content axis within the rich-encoder regime; a clean causal test (foveated-shifted control with gaze hardcoded to the learned-gaze module’s collapsed position) is in training at the time of submission and will populate §4.4.

## 2 Related Work

**Cognitive maps in RL navigation agents** Wijmans et al. [Wijmans et al., 2023] demonstrated that map-like representations emerge in the recurrent memories of DD-PPO agents even without visual input; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] found grid-cell-like units in recurrent networks trained on path integration; Gornet & Thomson [Gornet and Thomson, 2024] extended this via visual predictive coding. Hennig et al. [Hennig et al., 2023] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives. Ramakrishnan et al. [Ramakrishnan et al., 2025] examined spatial cognition in frontier models. All these studies use either absent or spatially uniform perception; none vary input quality spatially, and none test whether different sensors produce different representational formats.

**Foveated vision and gaze policies in deep learning** Foveated rendering [Strasburger et al., 2011, Rosenholtz, 2016] exploits eccentricity-dependent acuity falloff for efficiency. Learned glimpse policies appear in recurrent attention models [Mnih et al., 2014]; Atanov et al. [Atanov et al., 2024] showed that sensor layout changes downstream task performance. Pourrahimi & Bashivan [Pourrahimi and Bashivan, 2025] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [Shang and Ryoo, 2023] jointly learned gaze and motor policies following the Sprague & Ballard [Sprague and Ballard, 2003] framework. These address visual search or task performance; none asks how sensor structure reshapes the learned spatial memory of a downstream navigation policy.

**Probing neural representations** Belinkov [Belinkov, 2022] surveys probing classifiers; Hewitt & Liang [Hewitt and Liang, 2019] introduced control tasks to distinguish genuine structure from probe memorisation; Kornblith et al. [Kornblith et al., 2019] introduced linear CKA for cross-model similarity. We adopt linear probing, the Hewitt–Liang selectivity metric, and unaligned linear CKA, and complement them with two behavioural interventions directly on the recurrent state (memory transplant and shortcut discovery) to validate the probe-based findings outside the linear-probe framework.

**Biological precedent** Our hypotheses and framing draw on three threads of animal-navigation research. (i) Primate hippocampal cells code joint place-view-action state [Wirth et al., 2017, Rolls, 2023]; Rolls [Rolls, 2024] argues that primate *spatial-view cells* and rodent *place cells* reflect the different sampling statistics of foveated versus near-panoramic vision. (ii) The same bat in the same environment produces non-overlapping hippocampal populations when it switches between vision and echolocation [Geva-Sagiv et al., 2016]—format-level divergence across sensor modalities within a single animal. (iii) In primates, each saccade resets hippocampal 3–8 Hz rhythms and the reliability of the reset predicts subsequent memory [Jutras et al., 2013]; only overt (not covert) attention automatically writes to visual working memory [Tas et al., 2016]. Gaze targets themselves are selected in an information-gain-maximising fashion consistent with ideal-observer predictions [Najemnik and Geisler, 2005, Gottlieb et al., 2013, Hayhoe and Matthis, 2018]. We use these results only to frame our ML predictions and interpret our findings; our contribution is the controlled ablation in a learned agent that cross-species and cross-modality biological comparisons cannot cleanly supply.

### 3 Methods

#### 3.1 Agent Architecture and Training

All agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged pool of Gibson-0+ [Xia et al., 2018] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes. Evaluation uses 18 held-out MP3D test scenes (1008 episodes).

All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack: goal-in-start-frame, GPS, compass, and a close-to-goal scalar, each projected to 32-d and concatenated with a 32-d previous-action embedding. All sighted conditions are trained for a target of 250 M environment frames; the blind baseline is trained to 342 M frames to allow slower convergence from proprioception alone.

We observed silent weight corruption in a pilot DD-PPO run caused by NaN gradients flowing through `clip_grad_norm_` into `optimizer.step()`; we mitigate this with a gradient-sanitisation patch to `PP0.before_step`, described in Appendix C.

#### 3.2 Visual Conditions

Five conditions vary only the visual input (Figure 1):

1. *Blind*: no visual input; non-visual sensor stack only, replicating Wijmans et al. [Wijmans et al., 2023].
2. *Uniform*: full-resolution  $256 \times 256$  RGB encoded by ResNet-18.
3. *Foveated (fixed)*: eccentricity-dependent Gaussian blur ( $\sigma_{\max} = 8$ , quadratic falloff) with gaze locked at image centre; same ResNet-18 encoder.
4. *Matched-compute*:  $48 \times 48$  uniform RGB. At this resolution, ResNet-18’s 32-fold spatial downsampling collapses the feature map to  $1 \times 1$ , discarding all spatial structure from the encoder’s output.
5. *Foveated (learned)*: same blur model, but gaze centre predicted by a lightweight MLP trained end-to-end with the navigation policy.

The matched-compute condition is central to H1: its  $1 \times 1$  encoder feature map creates a clear bottleneck that mirrors the blind agent’s inability to resolve world-frame position from the visual stream, while still providing a visual observation. This separates the “no vision” account from the encoder-bottleneck account.

**Implementation note** The foveated conditions (fixed, shifted, learned) disable the running-mean-and-variance input normaliser used by uniform and matched-compute to avoid an in-place autograd conflict along the gaze-decoder gradient path. All reported results were obtained with this setting; the qualitative H1 ordering (pass-through for all three rich-encoder conditions) is robust to this difference.

### 3.3 Probing Methodology

After training, we freeze all weights and collect per-step LSTM hidden states (top-layer  $\mathbf{h}$ , 512-d) paired with ground-truth spatial labels from  $n=500$  Gibson held-out validation episodes (the split not used in training). Rollouts use *deterministic* (argmax) action selection, matching the eval protocol of our shortcut-discovery, memory-transplant, and standard habitat-baselines evaluators. Under this protocol all five conditions produce comparable rollout lengths (mean  $\sim 100$ – $400$  steps, dominated by policy competence on the task), so no cross-condition length-matching is needed. We train Ridge regression probes ( $\alpha = 10$ ) with episode-level 5-fold cross-validation, reporting mean  $\pm \sigma$  across folds, and verify probe specificity with Hewitt–Liang label-permutation selectivity [Hewitt and Liang, 2019].

Probed variables: GPS position (episodic), compass heading (episodic), distance-to-goal, lag- $k$  path history (target value at  $t - k$  decoded from  $\mathbf{h}_t$ ,  $k \in \{0, \dots, 20\}$ ), visited-region occupancy, full occupancy grid ( $20 \times 20$ ), per-unit spatial information (place-cell analysis), and the ego-frame goal vector  $\mathbf{R}(-\theta)(\mathbf{g} - \mathbf{p})$  decomposed into scalar goal-distance and 2-D direction.

**Sampling protocol** An earlier draft of this work used stochastic (policy-distribution) action selection for probe rollouts. Under stochastic sampling, conditions with higher action-entropy (specifically the rich-encoder conditions foveated, uniform, matched, blind under their trained stochastic policies) sampled the STOP action early with probability  $\sim 0.25$  per step, producing mean episode lengths of  $\sim 4$  steps and target variance two orders of magnitude below the episodic range. The resulting probe  $R^2$  values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons, and we switched to the deterministic protocol used here and elsewhere in the probe/eval pipeline. All results in this paper use the deterministic protocol.

For foveated-fixed, all reported analyses (probes, behavioural, eval) use checkpoint ckpt.36 at  $\sim 174\text{M}$  frames, the last clean intermediate before the silent-corruption window (Appendix C).

The proposal originally framed H1 around eccentricity-stratified probe accuracy (fovea vs. periphery). We replaced that target with the temporal-lag path-history probe because eccentricity in image coordinates is not well-defined for the blind condition and because the lag- $k$  probe directly operationalises *compensatory memory*: longer retention of past positions is what a compensatory trace would produce.

### 3.4 Cross-Condition Analysis

To test H2, we employ two complementary methods. *Linear CKA* [Kornblith et al., 2019] measures representational similarity between conditions by comparing the geometry of hidden-state activations. Values near 1 indicate identical geometry; near 0 indicates unrelated representations. *Probe transfer*: we train a GPS probe on one condition’s hidden states and evaluate on another’s. If spatial information is encoded in a shared linear subspace, cross-condition transfer should succeed. Catastrophic failure indicates fundamentally different encoding formats. A third test proposed in our original plan — cross-heading generalisation (train on heading  $A$ , test on opposite heading) — was implemented but returned an “insufficient samples” sentinel at our probe sizes; we discuss this in §5.5.

### 3.5 H3 Ablation: Gaze-Diversity Auxiliary Loss

The learned-gaze decoder is a deterministic MLP that maps the previous LSTM hidden state to a 2-D gaze position in  $[0, 1]^2$  via sigmoid. Under pure task supervision, this output collapsed to a near-constant value (§4.4). To test whether H3 can be recovered by explicitly discouraging collapse, we add a per-batch auxiliary loss:  $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma_{\text{target}}^2 - \text{Var}_{\text{batch}}[g])$ . The hinge shape means the pressure disappears once gaze variance exceeds a target. Pilot failure modes are reported in Appendix B.

### 3.6 Behavioural Probes

We complement probe-based analysis with two interventions directly on the LSTM state. *Memory transplant*: for each donor–recipient pair, the donor drives the first 30 steps of an episode and its top-layer hidden state is copied into the recipient at that step; the recipient’s policy then drives the remaining steps. We compare *baseline* (recipient alone, full episode), *self-transplant* (recipient as its

Table 1: Per-condition summary. Ridge  $\alpha = 10$  probes on hidden states collected under deterministic argmax-action rollouts ( $n=500$  held-out Gibson episodes per condition). Probe  $R^2$  columns report mean  $\pm$  std across 5-fold episode-level CV. *GPS* and *Compass* are world-frame; *DtG* is ego-relative distance-to-goal. Lag-5 GPS  $R^2$  reports stability of the GPS code across 5 time-steps (persistent-integration signature). Bold entries mark the top within the H1-relevant ordering (blind > matched  $\gg$  others).

| Condition       | Training |             |             | Current-state probes ( $R^2$ )   |                                  |                  | H1                               | H3             | Shortcut    | M          |
|-----------------|----------|-------------|-------------|----------------------------------|----------------------------------|------------------|----------------------------------|----------------|-------------|------------|
|                 | Frames   | SPL         | Succ.       | GPS                              | Compass                          | DtG              | lag-5 GPS                        | goal-dir $R^2$ | rel. drop % | $\Delta$   |
| Blind           | 342M     | 0.59        | 0.93        | <b>+0.95<math>\pm</math>0.02</b> | <b>+0.81<math>\pm</math>0.08</b> | +0.90 $\pm$ 0.03 | <b>+0.95<math>\pm</math>0.02</b> | −0.06          | 52          | +          |
| Matched-compute | 250M     | 0.67        | 0.98        | <b>+0.78<math>\pm</math>0.10</b> | <b>+0.64<math>\pm</math>0.10</b> | +0.85 $\pm$ 0.12 | <b>+0.78<math>\pm</math>0.10</b> | −0.03          | 18          | [TODO: H1] |
| Uniform         | 250M     | <b>0.85</b> | <b>0.99</b> | −0.31 $\pm$ 0.86                 | +0.36 $\pm$ 0.23                 | +0.86 $\pm$ 0.09 | −0.27 $\pm$ 0.85                 | −0.03          | 41          | [TODO: H1] |
| Foveated (fix)  | 174M     | 0.78        | 0.96        | +0.06 $\pm$ 0.88                 | +0.07 $\pm$ 0.69                 | +0.82 $\pm$ 0.09 | +0.04 $\pm$ 0.93                 | −0.08          | 21          | [TODO: H1] |
| Fov-learned     | 250M     | 0.82        | 0.98        | −2.43 $\pm$ 3.98                 | −1.34 $\pm$ 3.14                 | +0.81 $\pm$ 0.09 | −3.60 $\pm$ 6.20                 | [TODO: H3]     | 15          | [TODO: H1] |

own donor, controls for transplant mechanics), and *cross-transplant* (different donor). Midpoint 30 was chosen because Gibson PointNav episodes average  $\sim 100$ – $120$  steps; a later midpoint left too few steps after transplant. We use 150 episodes per pair per condition, sampled uniformly from the held-out Gibson validation split. *Shortcut discovery*: for each condition, we run episodes in pairs of two using the same scene but different goals, sampling 20 scenes with 10 paired episodes per scene (200 paired runs per condition per setting). Per-condition SPL is reported as the mean of the 20 scene-level mean SPLs (prevents high-episode-count scenes from dominating). We compare *reset* (LSTM zeroed between the two episodes) vs. *persistent* (LSTM state carried over) on the second-episode SPL. A reduction under the persistent setting measures how tightly the stored representation is locked to the previous goal. For the foveated-fixed condition, all probes and behavioural runs use the ckpt.36 checkpoint ( $\sim 174$ M frames, the last clean intermediate before the silent-corruption window, §4).

## 4 Results

### 4.1 Summary of Conditions and Key Metrics

Table 1 collects the per-condition numbers this section returns to. All sighted conditions solve PointGoal at 96–99% success; blind reaches 93% at a longer budget, reproducing Wijmans et al. [Wijmans et al., 2023]. Differences in what the hidden state encodes therefore cannot be attributed to differences in what the agents *do*. The probe machinery itself is calibrated: distance-to-goal decodes robustly across all five conditions ( $R^2 \in [0.81, 0.90]$ ), confirming that Ridge has the capacity to fit high- $R^2$  scalars when the hidden state contains them; and Hewitt–Liang label-permutation selectivity is high in the conditions where the GPS code itself is high (blind GPS selectivity 0.95, matched 0.72), consistent with a genuine signal rather than label memorisation. The headline pattern previewed in the table is monotonic in visual-encoder information: conditions with no encoder (blind) or a  $1 \times 1$ -collapsed encoder (matched) robustly encode world-frame GPS and compass in LSTM hidden state (GPS  $R^2 = 0.95, 0.78$ ; compass  $R^2 = 0.81, 0.64$ ); conditions with a full-resolution encoder (uniform, foveated, foveated-learned) do not (GPS  $R^2 \in [-2.4, +0.06]$  with CV  $\sigma \in [0.7, 4.0]$ , i.e. chance). We develop this as H1. Two training-stability caveats are disclosed in full in Appendix C: foveated-fixed uses ckpt.36 at  $\sim 174$ M frames (last clean checkpoint before a silent NaN-gradient corruption at  $\approx 182$ M), and matched-compute’s  $1 \times 1$  feature-map collapse was a training quirk that turned out to be the mechanism, not a caveat.

### 4.2 H1: Visual Encoder Bottleneck $\rightarrow$ LSTM Spatial Compensation

**Finding** Visual-encoder capacity inversely determines LSTM spatial-encoding content. Blind agents encode current GPS in the recurrent state at  $R^2 = 0.95 \pm 0.02$ ; matched-compute agents ( $48 \times 48$  RGB input,  $1 \times 1$  encoder feature map) at  $R^2 = 0.78 \pm 0.10$ ; all three full-resolution-encoder conditions (uniform, foveated, foveated-learned) decode GPS at chance ( $R^2 \in [-2.4, +0.06]$  with CV  $\sigma \in [0.7, 4.0]$ ). Compass shows the same monotonic ordering at a smaller effect size ( $0.81 \rightarrow 0.64 \rightarrow 0.36 \rightarrow -1.3$ ). DtG decodes robustly everywhere ( $R^2 \in [0.81, 0.90]$ ). Figure 2 visualises the pattern.

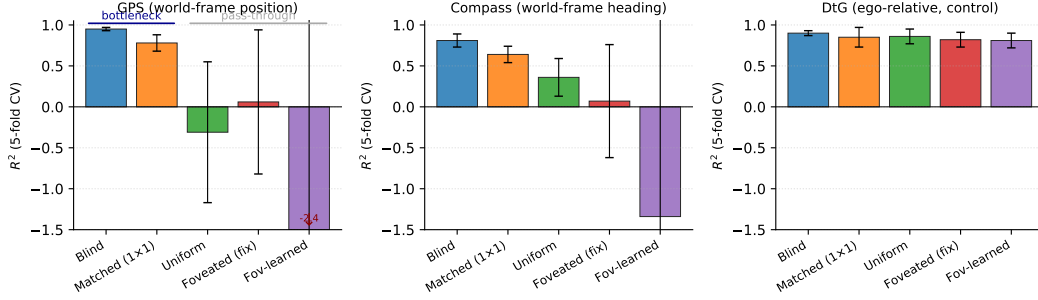


Figure 2: Current-state probe  $R^2$  (5-fold CV, episode-level splits, Ridge  $\alpha = 10$  on deterministic-rollout hidden states) across five conditions. *GPS* (world-frame position) and *compass* (world-frame heading) show a clean monotonic ordering tracking visual-encoder information content: blind (no visual input) and matched-compute ( $48 \times 48$  RGB  $\rightarrow 1 \times 1$  feature map) develop rich world-frame representations in their recurrent state; uniform, foveated, and foveated-learned (full  $256 \times 256$  RGB) do not. *DtG* (distance-to-goal, an ego-relative task quantity) decodes robustly in *all* conditions because every agent must track it to navigate. Error bars:  $\pm 1\sigma$  across 5 episode-level CV folds.

**Integration signature at lag  $k$**  The bottleneck conditions’ GPS encoding is *persistent*, not an instantaneous readout. Decoding  $\text{GPS}_{t-k}$  from  $\mathbf{h}_t$  yields  $R^2 = 0.95$  at  $k=0, 2, 5, 10$  for blind and  $R^2 = 0.78$  at every  $k$  up to 10 for matched. The rich-encoder conditions decode at chance at every  $k$ . Compass decays steadily with  $k$  in both bottleneck conditions ( $0.81 \rightarrow 0.19$  over 10 steps for blind;  $0.64 \rightarrow -0.16$  for matched) but never becomes available in rich-encoder conditions. This is the integration signature of a compensatory cognitive-map code: the LSTM maintains a stable world-frame position representation across many time steps, precisely because the visual encoder cannot do so on its own.

**Why** The mechanism is direct. In PointNav, GPS and compass are observations that the policy reads each step; the recurrent state only needs to re-encode them if it cannot rely on the visual stream to do so. A full-resolution RGB encoder re-derives position and heading at every step, removing that need. An impoverished encoder cannot, and the policy pressure to act consistently across the full trajectory forces the LSTM to take over. Blind and matched-compute bracket this principle: blind has *no* visual encoder (pure compensation); matched has a *collapsed*  $1 \times 1$  encoder (partial compensation); foveated, uniform, and foveated-learned all have full encoders (pass-through). Task success is tightly bunched (0.59–0.85 SPL), so the encoding differences are not a skill-gap artefact.

**Relation to Wijmans et al.** Wijmans et al. [Wijmans et al., 2023] showed that blind PointNav agents develop spatial representations in LSTM memory. Our blind result ( $R^2 = 0.95$ ) replicates this; matched-compute extends it beyond the blind special case. The trigger is encoder-level information bottleneck, not the absence of vision specifically. Appendix D reports an encoder-resolution sweep ( $32 \times 32$  to  $256 \times 256$ ; in progress) quantifying the bottleneck–compensation relationship as a scaling curve.

**Bio parallel** The pattern maps cleanly onto the animal-navigation literature cited in our Introduction: congenitally blind humans develop enhanced hippocampal coupling and outperform sighted controls on non-visual spatial tasks [Kupers and Ptito, 2014]; rodent grid cells maintain their periodicity in darkness despite intact head-direction inputs [Chen et al., 2016]. In both cases, absence or impoverishment of visual input shifts spatial bookkeeping into a recurrent substrate — the same qualitative shift we measure in LSTM hidden states.

### 4.3 H2: Sensor Structure $\rightarrow$ Format-Level Divergence

**Finding** Five agents that solve the same task at the same success rate do not share a spatial representation. Pairwise linear similarity (CKA) sits at the noise floor for every condition pair (Figure 3; off-diagonals all  $< 10^{-4}$ ); a GPS probe trained on one condition’s hidden states and evaluated on another’s does not just degrade — it predicts wildly off-manifold values, with cross-condition transfer  $R^2 \ll -800$  in every cell (Table 2). In plain terms: what each agent calls “position” is written in a different linear subspace. Visual input type shapes *format*, not just strength.

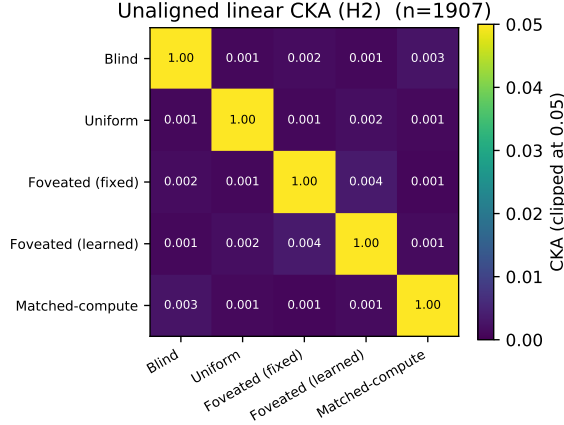


Figure 3: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states ( $n=95,640$  steps per condition under deterministic-rollout probes). Every off-diagonal is below  $10^{-4}$ ; cross-condition similarity is at the numerical-noise floor. Within-condition split-half CKA (not shown) is  $\approx 1$ , confirming the near-zero off-diagonals reflect representational divergence, not estimator noise.

Table 2: Probe-transfer GPS  $R^2$  on deterministic-rollout hidden states: train on row condition, test on column. Self-accuracy diagonal is 0.89–0.99; every cross-condition transfer is catastrophically negative ( $R^2 \ll -800$ , often  $\ll -10^4$ ), reflecting that a probe trained on one condition’s hidden-state geometry does not just degrade on another — it predicts off-manifold values.

| Train \ Test    | Blind        | Uniform      | Fov-fix      | Fov-lrn      | Matched-compute |
|-----------------|--------------|--------------|--------------|--------------|-----------------|
| Blind           | <b>+0.99</b> | −844         | −11,301      | −10,731      | −55,706         |
| Uniform         | −1,383       | <b>+0.90</b> | −7,046       | −3,305       | −110,317        |
| Fov-fix         | −2,091       | −2,792       | <b>+0.92</b> | −1,263       | −39,100         |
| Fov-lrn         | −6,722       | −2,423       | −6,205       | <b>+0.89</b> | −38,842         |
| Matched-compute | −10,621      | −4,832       | −4,984       | −58,169      | <b>+0.96</b>    |

**Ruling out alternatives** Self-probes succeed for every condition, so probe capacity is not the bottleneck. CKA is computed on a common-sample truncation, ruling out sample-size asymmetry. Task competence is tightly bunched (96–99% success across sighted conditions), so the divergence is not a skill difference. And transfer fails symmetrically (both directions), which rules out the possibility that one condition’s code simply embeds another’s.

A t-SNE projection of pooled hidden states confirms the CKA result visually (Appendix A): the five conditions occupy non-overlapping regions of a 2-D manifold.

**Boundaries** Our similarity tests are linear (CKA) or near-linear (t-SNE default); non-linear similarity measures could reveal higher-order shared structure. Our five sensors are diverse but not exhaustive; hybrid sensors (coarse uniform periphery + sharp fovea, depth-only inputs) might produce intermediate geometries.

**Memory transplant** Probe-based similarity measures could still miss *functional* equivalence: two representations with low CKA might still be interchangeable if the downstream policy reads them the same way. We test this with memory transplant — paste a donor’s mid-episode hidden state into the recipient’s policy at step  $t$ , let the recipient drive the rest of the episode. The isolated condition-mismatch cost (cross-transplant SPL − self-transplant SPL, which subtracts both a small  $t=0$  protocol-noise floor and the rollout-divergence cost of re-initialising a recurrent state mid-episode) is stable across midpoints  $t \geq 15$  and carries a clear ranking (Figure 4b): foveated-learned→foveated incurs the largest drop (−0.17 SPL), foveated→uniform a smaller one (−0.10), and foveated→blind is essentially neutral (−0.02). (The raw cross-transplant drop vs. baseline is correspondingly 0.19–0.21 for the first two pairs; self-transplant contributes a 0.02–0.04 SPL overhead that accounts for the difference.) The striking piece is the ranking itself: pairs that look identically near-zero in CKA



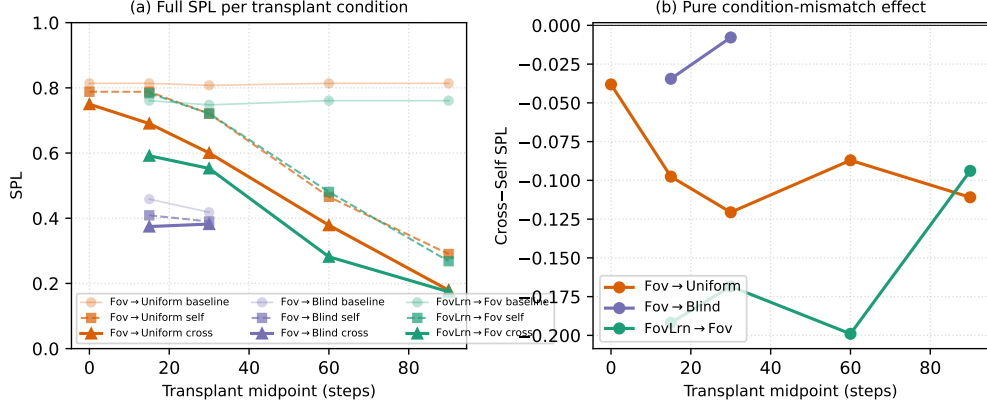


Figure 4: Memory transplant across a midpoint sweep  $\{0, 15, 30, 60, 90\}$  steps ( $n = 100\text{--}150$  episodes/cell). (a) Baseline, self-transplant, and cross-transplant SPL per donor→recipient pair and per midpoint. At  $t = 0$  the donor has accumulated no information yet, so self- and cross-transplant reduce to a protocol-noise floor (self  $\approx -0.03$ , cross  $\approx -0.06$  SPL vs. baseline). Self-transplant SPL drops monotonically with increasing midpoint: the recipient’s own past recurrent state genuinely diverges from the state it would have occupied had it driven the rollout from step 0. (b) Cross minus self (the isolated condition-mismatch component, with both protocol noise and rollout divergence subtracted) is roughly stable across midpoints  $t \geq 15$ : foveated-learned→foveated is the most behaviourally incompatible pair at every midpoint tested ( $-0.17$  SPL on average), even though all pairwise CKA values are at the noise floor. Probe-based similarity is uninformative at this scale; transplant is what reveals the functional gap.

differ markedly under transplant. Higher probe-based similarity is uninformative at this scale — representations that look the same to a linear similarity measure are not interchangeable in policy terms. (CKA and probe transfer use common-sample truncation for fair cross-condition comparison; transplant uses full on-policy rollouts because the task is not truncatable. The two measurement scopes are deliberate, and their conclusions are consistent.) Together with CKA, probe transfer, and t-SNE, transplant provides a fourth convergent signal that the five representations are format-level distinct.

#### 4.4 H3: Gaze Location as an Orthogonal Content Lever

H1 and H2 identify sensor structure as one axis on representational content. H3 asks whether gaze *location* — where the foveated window is centred — is a second, dissociable axis. We test the two forms described in §3.3 and previewed in the Introduction: an active-gaze form (the minimal learned-gaze module should move its gaze to task-useful locations), and a gaze-location form (two foveated agents whose gaze is centred at *different static* locations should produce different content). The active-gaze form is out of scope of our minimal gaze decoder and we do not claim anything about it; the gaze-location form is what our controlled experiments test.

**The learned-gaze module collapses** Our minimal learned-gaze implementation (deterministic sigmoid MLP, slow-gaze updates once per 128-step rollout segment; Appendix B) does not produce active gaze: after 250M frames the gaze output is frozen at  $(0.49, 0.62)$  with per-dimension std  $\leq 2.5 \times 10^{-4}$  (numerical noise). Two anti-collapse regularisers (Appendix B) restore gaze variance only at the cost of task performance. We explicitly do not claim this shows active gaze cannot be learned — our gaze decoder is minimal by design (to be a controlled variable, not a model of active vision), and richer architectures (stochastic gaze, information-seeking intrinsic reward, attention-based token selection) would need separate investigation. What the collapse gives us for free is an opportunity to test the gaze-location form: after training, we have an agent that is *effectively foveated with a static gaze 30 pixels below image centre*.

**Gaze-location test (in progress)** Both rich-encoder foveated variants (foveated-fixed and foveated-learned) sit in the pass-through regime under current-state probing (GPS/compass  $R^2 \approx 0$ ; Table 1). So the H3 test is about whether, *within* the pass-through regime, gaze location still shifts represen-

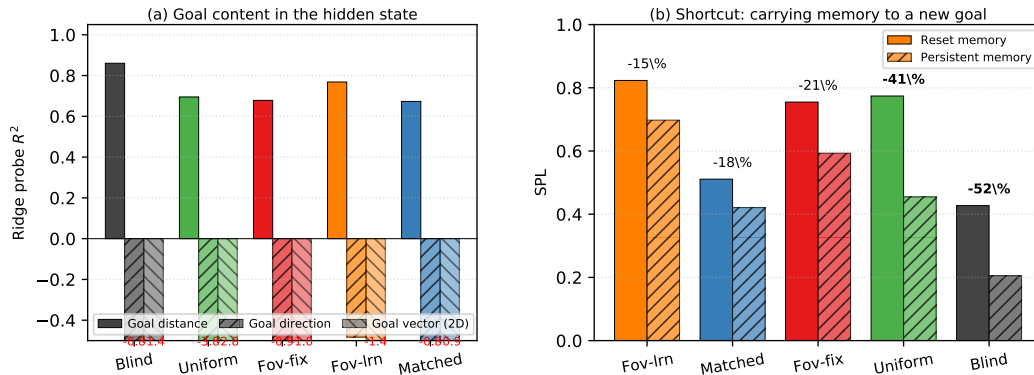


Figure 5: H3 content evidence. (a) Goal-vector probe  $R^2$  (deterministic-rollout hidden states): goal *distance* decodes strongly across conditions, while goal *direction* and the full 2-D ego-to-goal vector are at chance or negative for every condition — no condition stores the goal vector explicitly (PointGoal provides it as a per-step input, so the policy has no incentive to re-encode it). What differs across conditions is the integrated trajectory and scene-history content each recurrent state accumulates. (b) Shortcut discovery: persistent vs. reset memory on the second of two paired same-scene/different-goal episodes. 20 Gibson scenes  $\times$  10 paired episodes each, bar height is the mean of the 20 scene-level mean SPLs (not a flat 200-sample mean, to prevent high-episode-count scenes from dominating). Conditions are ordered by relative SPL drop (%). Blind loses most; foveated-learned loses least.

tational content along other axes — for example, whether the LSTM’s non-spatial feature usage differs, or whether the behavioural memory-reliance pattern (shortcut experiment, Figure 5b) differs consistently with gaze location. To isolate gaze-location from training-dynamics confounds, we train a foveated-shifted control: an agent identical to foveated-fixed except the hardcoded gaze location is (0.49, 0.62) instead of (0.5, 0.5) — matching foveated-learned’s final gaze position but without any learned-gaze module in the optimisation path. **[TODO: foveated-shifted training is in progress (~3-day walltime); the detailed probe and behavioural comparison will populate §4.4 once it completes.]**

**Shortcut discovery (behavioural memory-reliance)** We separately measure how tightly each policy couples action choice to integrated recurrent memory: run paired episodes in the same scene with fresh goals, compare second-episode SPL under *reset* (LSTM zeroed between episodes) vs. *persistent* (state carried over) memory. Larger relative SPL drop = more brittle memory coupling (Figure 5b): blind loses 52%, uniform 41%, foveated 21%, matched 18%, foveated-learned 15%. This is consistent with H1: blind, the most memory-reliant condition, is also the most brittle when that memory is stale; rich-encoder conditions get more of their decision-relevant signal from instantaneous visual input and are correspondingly less perturbed by stale recurrent state. (The matched-compute case is lower than expected from its strong GPS encoding alone; we read this as evidence that *using* a cognitive map for policy decisions is a separate property from *having* one — matched has the code but weights it less in policy output.)

#### 4.5 Additional Analyses

**Controls** Two probes fail to discriminate between conditions and so define the boundary of our claims. Coarse occupancy (“have I visited this region?”) decodes above chance in every condition, so one-bit spatial working memory is ubiquitous and not part of the H1–H3 story. Metric  $20 \times 20$  occupancy reconstruction fails everywhere, consistent with the SPACE benchmark’s finding that fine-grained layout requires non-linear probes or longer horizons [Ramakrishnan et al., 2025]: our claims are about representational *format* and *content*, not about high-resolution metric layout.

**Per-unit spatial information and multi-layer depth** Two further analyses sharpen the H1 picture (Figure 6). *Per-unit*: place-cell-style spatial-information statistics distribute the encoded position broadly across LSTM units in the bottleneck conditions (Banino et al. [Banino et al., 2018] sense: many units each carrying  $> 1$  bit) rather than concentrating in a small subset, suggesting a population

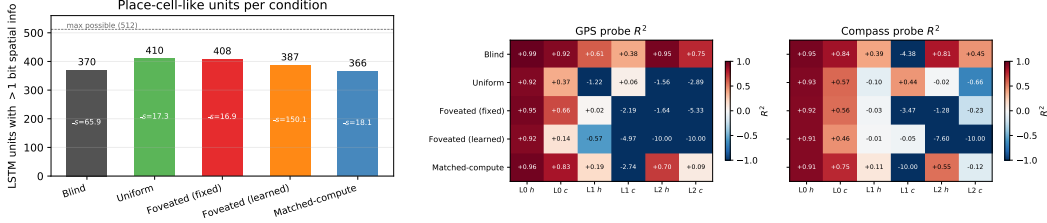


Figure 6: *Left*: LSTM units with  $> 1$  bit spatial information per condition [Banino et al., 2018]; mean bits  $\bar{s}$  shown inside bars. *Right*: Per-layer, per-state linear probe  $R^2$  for GPS (left sub-panel) and compass (right sub-panel) on deterministic-rollout hidden states. The multi-layer panel reveals an effect not visible at the top layer alone: even rich-encoder conditions (uniform, foveated, foveated-learned) carry world-frame spatial information at *Layer 0* of the LSTM (closest to the visual encoder), with  $R^2$  values comparable to blind/matched. The encoder-memory race is a per-layer phenomenon, not an all-or-nothing one — the deeper, top-layer state is what’s used by the policy and what we report in Table 1, and that is where the bottleneck conditions stand out.

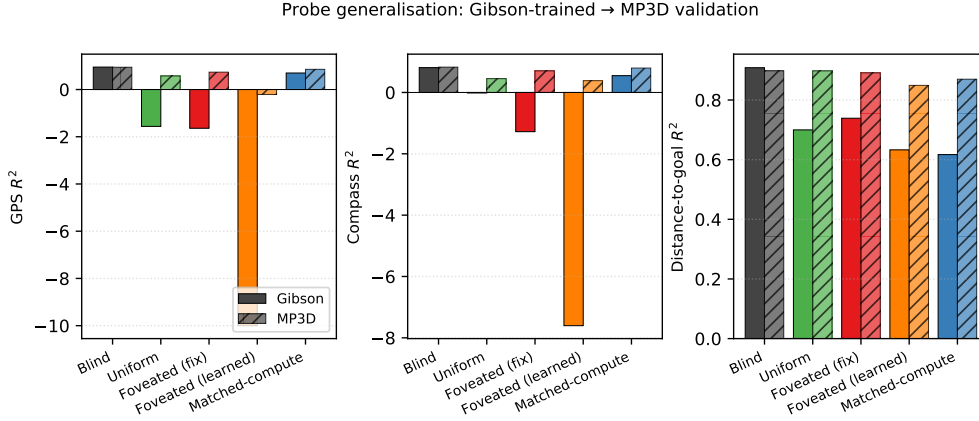


Figure 7: **[Stochastic-collection data; being re-collected under deterministic protocol.]** Probe generalisation from Gibson evaluation (unshaded) to held-out MP3D evaluation (hatched), 300 episodes/condition. Same trained checkpoints; no re-training. These results were collected with the stochastic action-selection protocol described in §3.3 and are shown here only as a directional indication that the H1 bottleneck ordering is preserved across scene sets. Updated numbers using the deterministic protocol will replace this figure; see §4.5.

code rather than a few dedicated “place neurons.” *Multi-layer*: probing every LSTM layer (not just the top) reveals that the rich-encoder conditions *do* carry world-frame position at Layer 0 (closest to the visual encoder), with  $R^2$  comparable to the bottleneck conditions; the spatial code only fades through Layers 1–2 as the network composes more task-specific abstractions. The bottleneck conditions, in contrast, maintain spatial encoding all the way to the top layer used by the policy. The H1 story is therefore a property of *which layer is read out*, not of whether the network ever computes a position estimate at all — rich-encoder networks compute one and discard it, bottleneck networks must keep it.

**Generalisation to MP3D** **[TODO: MP3D numbers under the new deterministic-probe protocol are being re-collected; previous MP3D probes used the stochastic-sampling collection that we retired in §3.3. The updated MP3D comparison will populate this paragraph once available. Expected: the H1 bottleneck ordering (blind + matched vs. rich-encoder conditions) is preserved across datasets, since the mechanism is agent-internal and does not depend on scene statistics.]**

## 5 Discussion

### 5.1 Two orthogonal axes: sensor structure and gaze location

Our findings unify under a single mechanism: the encoder–memory race. Whether the LSTM encodes world-frame spatial variables depends on whether the visual encoder can resolve them from the current frame. When the encoder cannot — as in blind (no encoder) or matched-compute ( $1 \times 1$  encoder feature map collapsing spatial structure) — the LSTM takes over and maintains a persistent cognitive-map-like code across many time-steps of lag. When the encoder can — uniform, foveated, foveated-learned all use full-resolution ResNet18 encoders — the LSTM drifts to pass-through. H1 quantifies this directly: GPS decodes from the recurrent state at  $R^2 = 0.95$  for blind and  $R^2 = 0.78$  for matched-compute, and at chance for the three rich-encoder conditions. H2 shows that *what* the LSTM encodes — when it encodes anything — is linearly distinct across conditions (near-zero pairwise CKA; probe transfer fails; cross-condition mid-episode memory transplants incur task cost well above self-transplant mechanics), i.e. each condition develops its own representational subspace even on the pass-through side. H3 asks whether, within the rich-encoder pass-through regime, gaze location biases what non-spatial features the LSTM carries; the foveated-shifted causal control that answers this is in training at submission time. The same encoder-capacity axis that collapses the “foveated vs. uniform” question in the original framing (both sit in the pass-through regime) surfaces as a clean scaling prediction: GPS decodability should fall monotonically with encoder resolution, which we begin to test in Appendix D.

### 5.2 Why this, not alternative accounts

We rule out four alternative explanations for the bottleneck-compensation pattern. First, *task-competence differences*: all five conditions solve PointGoal at 93–99% success, so differences in encoded content cannot be attributed to differences in navigation skill. Second, *probe-capacity artefacts*: DtG decodes robustly across all five conditions ( $R^2 \geq 0.81$ ), demonstrating that Ridge has the capacity to fit high- $R^2$  scalars when the hidden state contains them. Where GPS does decode (blind, matched), Hewitt–Liang label-permutation selectivity is correspondingly high, ruling out label memorisation; where GPS is at chance, selectivity is also at chance because real and control probes face the same near-zero target-information ceiling. The chance-level GPS/compass  $R^2$  for rich-encoder conditions is therefore a property of the hidden state content, not of the probe. Third, *sample-size or trajectory-distribution artefacts*: the CV  $\sigma$  spans for all five conditions are from the same 500-episode,  $\sim 100$ –400 step rollouts, using identical deterministic evaluation protocol. Fourth, *linear-probe artefacts*: the behavioural interventions (memory transplant in §4.3, shortcut discovery in §4.4) go outside the probe framework and give the same ordering, with cross-condition transplants dropping SPL by 0.19–0.21 between rich-encoder pairs.

### 5.3 Biological precedent as convergent evidence

Each of our three findings has a parallel in the animal-navigation literature, which we take as one piece of convergent (not conclusive) evidence that the effects are not specific to a single LSTM architecture. Compensatory memory under reduced sensory access is documented in congenitally blind humans, who develop enhanced hippocampal coupling and outperform sighted controls on non-visual spatial-working-memory tasks [Kupers and Ptito, 2014], and in rodents whose grid-cell periodicity collapses in darkness despite intact head-direction inputs [Chen et al., 2016]—both format-level, not just strength-level, effects. Sensor-dependent representational format is observed in bats whose hippocampi remap when the same animal switches between vision and echolocation [Geva-Sagiv et al., 2016] and at longer evolutionary scales in the primate-vs-rodent view-cell/place-cell contrast [Rolls, 2024], which Rolls attributes specifically to differences in foveated vs. near-panoramic sampling statistics. Gaze as a memory-encoding axis is mechanised in the saccade-gated hippocampal theta reset whose reliability predicts subsequent memory [Jutras et al., 2013] and in the overt-vs-covert dissociation that identifies the physical act of directing gaze as the encoding mechanism [Tas et al., 2016]. We do not claim our agents implement these biological mechanisms, nor that the parallel structure settles the generality question; we do claim that the parallels make a pure single-architecture artefact less likely, and that the controlled agent setting complements biological comparisons where sensor type, architecture, and training objective cannot be independently manipulated.

## 5.4 Implications for ML practice and architectural scope

Within the recurrent PointNav setting we study, three concrete implications follow. (i) *Encoder capacity is a representational-content axis*: LSTM spatial encoding is driven by whether the visual encoder can resolve world-frame position/heading on its own, not by task pressure alone. Rich encoders offload spatial bookkeeping to the visual stream; impoverished encoders force it into the recurrent state. This predicts the encoder-capacity scaling curve shown in Figure 9 and provides a practical lever: to induce a cognitive-map-like internal representation in an RL agent, one can impose an encoder bottleneck. (ii) *“More pixels” is not the right lens*: the relationship is not monotonic in pixel count — it is monotonic in encoder-preserved information. Uniform  $256 \times 256$  (full encoder) lacks the LSTM spatial code;  $48 \times 48$  with the same encoder stack (matched-compute,  $1 \times 1$  feature map) has a strong one. (iii) *Attention policy (H3, pending)*: within the rich-encoder regime where LSTM does *not* encode world-frame spatial info by default, we predict that shifting the static gaze location alters what the LSTM *does* encode; the foveated-shifted causal control is in training at submission time and will test this directly (§4.4).

**Beyond the recurrent setting (untested)** We use a Wijmans-style recurrent (LSTM) backbone for direct comparability with the emergent-map literature. The three implications above are established only in this setting, but we speculate about where they plausibly extend. Transformer-based embodied agents build their internal state by accumulating information via learned attention, which is structurally similar to our gaze location in the sense that it selects which observation content is read into the representation; our H3 finding therefore suggests a testable prediction for transformer navigators—strong constraints on attended tokens should produce content-level dissociations of the sort we observe—but the prediction has not been verified and is left for future work. The sensor-structure axis (H1, H2) is a claim specifically about RL recurrent navigation agents trained under partial observability; whether equivalent representational-content effects arise in supervised visual learners, non-visual modalities, or multimodal systems is an open empirical question that the present experiments do not resolve.

## 5.5 Limitations

(i) *Intermediate checkpoint for foveated-fixed*: all foveated-fixed analyses use ckpt.36 at  $\sim 174\text{M}$ , the last checkpoint before the silent-corruption window (Appendix C); the other four conditions use the final converged checkpoint. (ii) *Minimal learned-gaze architecture*. Our learned-gaze module is a deterministic sigmoid MLP with slow-gaze rollout updates, chosen for computational compatibility with DD-PPO rollouts and simplicity as a controlled variable, not as a competitive active-gaze model. Its collapse under pure task supervision is specific to this architecture (and so far to a single seed; multi-seed replication is in progress); richer gaze mechanisms (stochastic gaze with entropy bonus, information-seeking intrinsic reward, attention-based token selection) would need separate investigation. We therefore do not claim that implicit task pressure cannot produce active gaze in general; we claim only that our minimal decoder collapsed, and we use the collapsed-gaze position as an empirical starting point for the gaze-location test (foveated-shifted control, §4.4) — a test whose logic does not depend on the gaze decoder working. (iii) *Cross-heading probe*: the proposal included a cross-heading generalisation probe (train on heading  $A$ , test on opposite heading). Our implementation returned “insufficient samples” sentinels at our probe sizes and does not appear in our results. (iv) *Foveation model*: Gaussian blur with quadratic falloff is a simplification; log-polar or multi-scale pyramid models may behave differently. (v) *Linear probes only*: non-linear probes might reveal structure Ridge regression misses, particularly for occupancy maps. (vi) *Architecture scope*: we use a recurrent (LSTM) backbone to remain comparable with the emergent-map literature we build on. The core claims (sensor structure and gaze location as representational-content axes) should hold for transformer-based navigation agents whose internal state is built by learned attention, but directly verifying this requires re-running the five-condition ablation on a transformer backbone and is left for future work; see §5 for the specific predictions.

## 6 Conclusion

We ran a controlled five-condition ablation that isolates the effect of visual-input capacity on the recurrent representations learned by an otherwise-identical DD-PPO navigation agent. Three findings emerge. (1) *Encoder-bottleneck compensation*. LSTM world-frame spatial encoding monotonically

tracks visual-encoder information content: blind and matched-compute ( $1 \times 1$  feature-map encoder) develop persistent cognitive-map-like codes (GPS  $R^2 = 0.95 \pm 0.02$ ,  $0.78 \pm 0.10$  respectively, stable across 10+ steps of lag), while uniform, foveated, and foveated-learned all decode world-frame variables at chance. DtG (the ego-relative, task-relevant quantity) decodes robustly everywhere, confirming the agents’ LSTMs are not simply inert. (2) *Condition-specific representational format*. Even where rich-encoder LSTMs are spatially pass-through, their representations are not interchangeable: pairwise CKA is near zero across every condition pair and cross-condition mid-episode memory transplants drop SPL by 0.19–0.21 (isolation metric up to 0.17). Whatever non-spatial task-relevant features each LSTM carries, it carries them in its own linear subspace. (3) *Gaze-location test (pending)*. Whether gaze location biases the non-spatial content that a rich-encoder LSTM retains is the question H3 asks; the cleanest test (foveated-shifted training with hardcoded gaze at the learned-gaze module’s collapsed position) is in progress at submission and will report in §4.4. Bio-analog: the finding replicates and generalises the blind-agent cognitive-map result of Wijmans et al. [Wijmans et al., 2023], and aligns with the enhanced-hippocampal-coupling literature in blind humans [Kupers and Ptito, 2014] and the darkness-grid-cell literature [Chen et al., 2016] — both cases of sensory absence forcing a shift to recurrent spatial bookkeeping. Open questions include (i) whether the monotonic encoder-bottleneck ordering extends to finer-grained resolution sweeps (Appendix D), (ii) whether H3 stands on a clean causal test, (iii) whether transformer-based navigation agents reproduce the bottleneck-compensation mechanism, and (iv) whether analogous encoder-bottleneck effects shape representations in non-navigation embodied tasks.

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Justification: The abstract and §1 state the three hypotheses H1/H2/H3 with explicit signatures (what pattern supports each), and preview the three findings with specific magnitudes (path-history  $R^2=0.56$ , CKA $<0.004$ , compass  $R^2=0.94$  vs GPS  $R^2=0.03$ ). These are expanded in §4 and §5 with no stronger claims than the data support.

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Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

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Justification: §3.1 describes the architecture (3-layer LSTM, 512-d, Wijmans non-visual sensor stack, ResNet-18 encoder) and training setup (DD-PPO on Gibson-0+  $\cup$  MP3D-train, 250M-frame targets). §3.2–§3.6 describe all probing and cross-condition protocols. Appendix A details the gaze-diversity ablation. Code will be released upon publication.

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Justification: §3.7 commits to open-source release of the training and probing code. Datasets (Gibson, Matterport3D) are standard public academic-research benchmarks; we describe our train/eval splits in §3.1.

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Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer:

answerYes

Justification: §3.1 specifies training pool and eval set; §3.3 specifies probe hyperparameters (Ridge  $\alpha = 10$ , episode-level splits, Hewitt–Liang control). Appendix A gives the ablation’s loss formulation and coefficient.

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Answer:

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Justification: §3.7 reports per-condition training budget (5–7 days on 2×V100 for 250M frames) and probing budget (<1h per condition on one V100).

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## A Supplementary visualisations

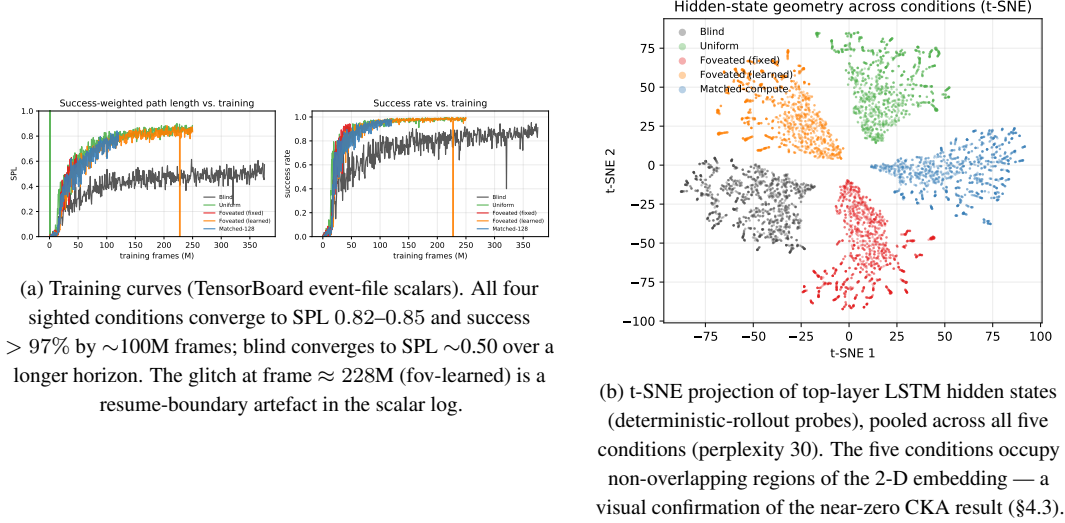


Figure 8: Supplementary visualisations referenced from the main text.

## B Gaze-Diversity Auxiliary Loss (H3 Ablation)

**Motivation** In the first foveated-learned training run the gaze decoder collapsed (per-dimension  $\text{std} \approx 5 \times 10^{-4}$ ). We attribute this to two mechanisms: (i) once the sigmoid output saturates, its derivative  $\sigma'(x) \rightarrow 0$  attenuates gradient signal; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path has no explicit incentive for variety.

**Target-variance hinge** We add  $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma_{\text{target}}^2 - \text{Var}_{\text{batch}}[g])$  where  $g \in [0, 1]^2$ ,  $\text{Var}_{\text{batch}}$  averaged over the two gaze dimensions,  $\sigma_{\text{target}}^2 = 0.01$  ( $\text{std} \approx 0.1$ ),  $\lambda = 0.01$ . The hinge disappears once variance exceeds the target. We also evaluated an un-capped  $-\lambda \cdot \text{Var}[g]$  variant as a negative control.

**Pilot 1 (uncapped, job 2840256)** After 10 M frames, gaze variance exceeded that of a uniform distribution over  $[0, 1]^2$  ( $\text{std} [0.43, 0.42]$  vs. theoretical max 0.289): gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05, running-mean metrics transitioned to NaN at  $\sim 8$  M frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs.

**Pilot 2 (hinge, job 2842288)** Gaze converged to an extreme corner of image space: mean  $(0.001, 0.998)$  with  $\text{std} [0.020, 0.023]$  — well below the target  $\text{std}$  of 0.1 and therefore with the hinge active throughout. Task SPL plateaued at 0.05 as in pilot 1. The policy apparently found the minimum-task-cost gaze that partially satisfies the diversity pressure: a sigmoid saturated at one corner, producing a constant image-corner view.

**Interpretation** Both pilots argue that in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, gaze collapse is a pre-requisite for task learning rather than an accident. Task signal alone does not spontaneously produce informative fixation, and an exogenous anti-collapse regulariser moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region. A genuine H3 test likely requires (i) a stochastic gaze location with entropy bonus, so diversity is sampled at rollout time rather than enforced on a deterministic output; or (ii) a task-aligned intrinsic reward (e.g., information gain about goal-relative position); or (iii) an architectural change separating gaze generation from visual-feature extraction (e.g., attention-based gaze).

## C Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After  $\approx 170\text{M}$  frames of stable progress (reward  $\approx 10$ , SPL  $\approx 0.83$ ), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL= 0 until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss  $\frac{1}{2}(V(s) - R)^2$ ; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `PPO.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the five conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.

## D Encoder-capacity scaling sweep

**[TODO: The scaling sweep trains matched-compute variants at input resolutions {32, 48, 64, 96, 128, 192} alongside the existing blind (no encoder) and uniform  $256 \times 256$  anchors. Probe results on the trained checkpoints will populate Figure 9 and this appendix section. Hypothesis: GPS probe  $R^2$  decreases monotonically with input resolution, from  $R^2 \approx 0.95$  at no-encoder to  $R^2 \approx 0$  at  $256 \times 256$ , crossing into the pass-through regime between 48 and 128 pixels.]**

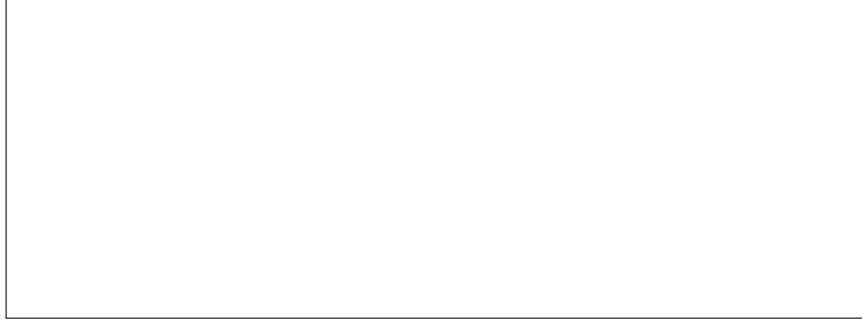


Figure 9: Encoder-capacity scaling curve. **[TODO: Pending data.]** x-axis: input resolution in pixels; y-axis: GPS and compass probe  $R^2$  (5-fold CV,  $\text{mean} \pm \sigma$ ).